T11

$H_0: \ \xi \sim p_0(x) = 1\{(0;1)\}$

$H_1: \ \xi \sim p_1(x) = \frac{e^{1-x}}{e-1}\{(0;1)\}$

а) $n=1$

$$l(x) = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq C \Rightarrow G: X \leq A \text{ — кр. область}$$

$$P(X \leq A \mid H_0) = \alpha$$

$$\int_0^A 1\,dx = A = \alpha \Rightarrow G: X \leq \alpha$$

$$W = P(X \leq A \mid H_1) = \int_0^A \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1}(1-e^{-\alpha}) \Rightarrow \begin{array}{l} \alpha_1 = P(X \leq A \mid H_0) = \alpha \\ \alpha_2 = P(X > A \mid H_1) = 1 - W = \\ = \frac{e^{1-\alpha}-1}{e-1} \end{array}$$

б) $n=2$

$$l = \frac{L_1}{L_0} = \frac{e^{1-x_1}e^{1-x_2}}{(e-1)^2} \geq C$$

$$\iint\limits_{\substack{x_1+x_2 \leq B \\ 0<x_1<1 \\ 0<x_2<1}} dx_1\,dx_2 = \frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha}$$

$$B \leq 1 \Rightarrow \alpha \leq \frac{1}{2}$$

$$G: X_1 + X_2 \leq B$$

$$P(X_1 + X_2 \leq B \mid H_0) = \alpha$$

$$W = P(X_1 + X_2 \leq B \mid H_1) = \iint\limits_{x_1+x_2 \leq B} \frac{e^{1-x_1}e^{1-x_2}}{(e-1)^2}\,dx_1\,dx_2 = \frac{e^2}{(e-1)^2}\int_0^B dx_1 \int_0^{B-x_1} e^{-x_1}e^{-x_2}\,dx_2 =$$

$$= \frac{e^2}{(e-1)^2}(1 - e^{-B} - Be^{-B})$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - W = 1 - \left(\frac{e^2}{(e-1)^2}\right)(1 - e^{-B} - Be^{-B})$$

$\alpha = 0{,}2$

$H_0: \xi \sim \frac{1}{3}\{4\} + \frac{1}{6}\{3\} + \frac{1}{4}\{1,2\}$

$H_1: \xi \sim \frac{1}{4}\{1,2,3,4\}$

Тройка выпала m_3 раз, четверка m_4 раз, соответственно 1 или 2 выпало $n - m_3 - m_4$ раз

$$l(\vec{X}_2) = \frac{L_1}{L_0} = \frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{3}\right)^{m_4}\left(\frac{1}{6}\right)^{m_3}\left(\frac{1}{4}\right)^{2-m_4-m_3}} = \frac{\left(\frac{1}{4}\right)^{m_4+m_3}}{\left(\frac{1}{3}\right)^{m_4}\left(\frac{1}{6}\right)^{m_3}}$$

(m_3, m_4)	$(0,0)$	$(0,1)$	$(0,2)$	$(1,0)$	$(1,1)$	$(2,0)$
l	1	$\frac{3}{4}$	$\frac{9}{16}$	$\frac{2}{3}$	$\frac{9}{8}$	$\frac{4}{9}$
$H_0 \to P$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{36}$
$H_1 \to P$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$

$$P(l \geq C \mid H_0) \leq \alpha = 0{,}2$$

1) $(2,0)$: $P = \frac{1}{36} \leq 0{,}2$ $\quad \frac{1}{36} + \frac{1}{6}$

2) $(1,0)$: $P = \frac{10}{36} \leq 0{,}2$ $\quad \frac{1}{36} + \frac{1}{6} + \frac{1}{9}$

3) $(1,1)$: $P = \frac{11}{36} > 0{,}2$

$C \quad m_3 \geq 1$

$\alpha = \frac{9}{36}$

$W = P(l \geq C \mid H_1) = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}$

$\alpha_2 = \frac{11}{16}$

ТПЧ $\xi = N(a, \sigma_x^2)$ $\eta = N(b, \sigma_y^2)$

$\sigma_x^2 = 2$ $\sigma_y = 1$

$X = \{-1.11, -6.10, 2.42\}$ $Y = \{-2.29, -2.91\}$

$H_0: a = b$

$H_1: a > b$

$\alpha = 0{,}05$

[redacted]

$$\Delta = \frac{\bar{x} - \bar{y} - a - b}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

$G: \Delta \geq u_{1-\alpha}$

$$\Delta \overset{H_0}{=} \frac{\bar{x} - \bar{y} -}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

$P(\Delta \geq u_{1-\alpha} | H_0) = \alpha$

$u_{\alpha-1} = 1{,}645$

$\tilde{\Delta} = 0{,}929 < u_{\alpha-1} \Rightarrow$ нет оснований отвергать H_0

$$W(a,b) = 1 - P\left(\frac{a-b}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} > c\right)$$

График отдельно

Т13.

$n = 139 \qquad m = 1000$

$\xi_1 \sim N(a, \varphi_1^2) \qquad \eta_1 \sim N(b, \psi_1^2)$

$\xi_2 \sim N(c, \varphi_2^2) \qquad \eta_2 \sim N(d, \psi_2^2)$

$S_{1n} = 5{,}722 \qquad S_{1m} = 6{,}161$

$S_{2n} = 4{,}612 \qquad S_{2m} = 5{,}055$

$H_0: \varphi_1^2 = \psi_1^2 \qquad \varphi_2^2 = \psi_2^2$

$H_1: \varphi_1^2 \neq \psi_1^2 \qquad \varphi_2^2 \neq \psi_2^2$

Если H_0 верна, то

$$\Delta = \frac{S_x^2}{S_y^2} \sim F(n-1, m-1)$$

$$P(|\Delta| \geq C) = \alpha \qquad \alpha = 0{,}05$$

$$C_1 = F_{\frac{\alpha}{2}}(138, 999)$$

$$C_2 = F_{1-\frac{\alpha}{2}}(138, 999)$$

$C_1 = 0{,}767 \qquad C_2 = 1{,}272$

$\frac{S_{1n}^2}{S_{1m}^2} = 0{,}86 \qquad \frac{S_{2n}^2}{S_{2m}^2} = 0{,}83 \Rightarrow$ нет оснований отвергнуть H_0

$$W(\varphi^2, \psi^2) = 1 - P\left(C_1 < \frac{\psi^2}{\varphi^2} < C_2\right)$$

c) Асимптотический критерий по выборке объема n: $\vec{X}_n$

$$\ell(\vec{X}_n) = \frac{L_1(\vec{X}_n)}{L_0(\vec{X}_n)} = \frac{\prod^n p_1(x_i)}{\prod^n p_0(x_i)} = \prod_i^n \frac{e^{1-x_i}}{e-1} \geq C$$

$$\sum_{i=1}^n (1-x_i) - n\ln(e-1) \geq \ln(C)$$

$$G: \sum_{i=1}^n x_i \leq A$$

$$\bar{X} \leq B$$

$$P(\bar{X} \leq B \mid H_0) = \alpha$$

УПТ: $\frac{\bar{X} - M\xi}{\sqrt{D\xi}}\sqrt{n} \sim N(0,1)$

Из $H_0 \to M\xi = \frac{1}{2}$; $D\xi = \frac{1}{12} \Rightarrow P(\bar{X} \leq B \mid H_0) = P\left(\frac{\bar{X} - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n} \leq \underbrace{\frac{B - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n}}_{u_\alpha}\right)$

$$\Rightarrow B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$\Rightarrow G: \bar{X} \leq \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{X} \leq B \mid H_1)$$

$$H_1 \Rightarrow M\xi = \int_0^1 \frac{xe^{1-x}}{e-1}\,dx = \frac{e-2}{e-1} \approx 0{,}418$$

$$M\xi^2 = \frac{e}{e-1}\int_0^1 x^2 e^{-x}\,dx = \frac{2e-5}{e-1}$$

$$D\xi = M\xi^2 - (M\xi)^2 = \frac{e^2 - 3e + 1}{(e-1)^2}$$

УПТ

$$W = P\left(\frac{\bar{X} - M\xi}{\sqrt{D\xi}}\sqrt{n} \leq \underbrace{\frac{B - M\xi}{\sqrt{D\xi}}\sqrt{n}}_{\gamma}\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,dx$$

$$\gamma = \frac{\frac{1}{2} - 0{,}418 + \frac{u_\alpha}{\sqrt{12n}}}{\sqrt{D\xi}}\sqrt{n} \xrightarrow{n \to \infty} +\infty \Rightarrow W \to 1$$

$\Rightarrow$ критерий сост.

$\alpha_1 = \alpha$

$\alpha_2 = 1 - W$

8) $X_{min} < C$

$\alpha_1 \le \alpha$

$\alpha_2 \to \min$

$$F(x) = \int_{-\infty}^{x} p_0(t)\,dt$$

$$\varphi(c) = \int_{-\infty}^{c} p_1(x)\,dx$$

$$\alpha_1 = P(X_{min} < c \mid H_0) = 1-(1-F(c))^n \le \alpha$$

$$F(c) \le 1-\sqrt[n]{1-\alpha}$$

$$\alpha_2 = P(X_{min} \ge c \mid H_1) = 1 - P(X_{min} < c \mid H_1) = 1-1+(1-\varphi(c))^n = (1-\varphi(c))^n \longrightarrow \min_c$$

$$\frac{\partial \alpha_2}{\partial c} = n(1-\varphi(c))^{n-1}\cdot(-p_1(c)) = \frac{-ne^{1-c}}{e-1}(1-\varphi(c)$$

$$\varphi(c) = \int_0^c p_1(x)\,dx = \frac{e}{e-1}\int_0^c e^{-x}\,dx = \frac{e(1-e^{-c})}{e-1}$$

$$1-\varphi(c) = \frac{e^{1-c}-1}{e-1}$$

$\frac{\partial \alpha_2}{\partial c} < 0$ $\quad c \in (0,1) \Rightarrow \min$ на $c \le 1-\sqrt[n]{1-\alpha}$ достиг при $c = 1-\sqrt[n]{1-\alpha}$

Итог: $G: X_{min} < 1-\sqrt[n]{1-\alpha}$

$$\alpha_2 = (1-\varphi(1-\sqrt[n]{1-\alpha}))^n = \left(\frac{e^{\sqrt[n]{1-\alpha}}-1}{e-1}\right)^n$$

$$\left(e^{\sqrt[n]{1-\alpha}} = e^{1+\frac{1}{n}\ln(1+\alpha)+o(\frac{1}{n})} = e\left(1+\frac{1}{n}\ln(1+\alpha)+o\left(\frac{1}{n}\right)\right)\right) *$$

$$\alpha_2 = \left(\frac{e\left(1+\frac{1}{n}\ln(1+\alpha)+o(\frac{1}{n})\right)-1}{e-1}\right)^n = \left(1+\frac{e\ln(1-\alpha)}{n(e-1)}+o\left(\frac{1}{n}\right)\right)^n *$$

$$\lim_{n\to\infty}\alpha_2 = (1-\alpha)^{\frac{e}{e-1}}$$

$$\lim_{n\to\infty} w = \lim_{n\to\infty}(1-\alpha_2) = 1-(1-\alpha)^{\frac{e}{e-1}} \ne 1$$

Критерий не явл. состоятельным

$$\alpha_1 = 1-(1-F(1-\sqrt[n]{1-\alpha}))^n = 1-(\sqrt[n]{1-\alpha})^n = \alpha$$